

GUIDE TO COMPETITIVE PROGRAMMING:

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It's advice to switch to CPP to avoid TLE issues at Codeforces and some platforms..
Learn C++ STL from codeverse or takeUforward channel ..

Follow CP sheet for practice problems:

CP SHEET: https://bit.ly/tuf_CPList

SDE SHEET: https://bit.ly/takeUforward_SDE

1. Pattern printing problems (Sourabh Shukla Videos on Youtube)
2. time complexity analysis (Don't stress too much, keep participating in contests, you will keep learning)
3. linear search and circular array representation (any article, no need to practice)
4. palindrome and other numbers(perfect, Armstrong) for basic number problems
5. Simple Hashing Problem(frequency counting and stuffs)
6. Prefix Sum Problems(1D and 2D) {CP Sheet has problems}
7. Sliding window technique(CP Sheet has problems)

Basics of number theory

1. Binary Search (TakeUforward Playlist and CP Sheet practice problems)
2. GCD of 2 numbers in logarithmic time(Euclidean and Extended Euclidean Algorithm) (CodeNCode)
3. linear Diophantine Equation (CodeNCode)
4. Checking prime in $\sqrt{n}$ complexity (Learn from Codeverse Channel)
5. Sieve of Eratosthenes(Learn from Codeverse Channel)
6. Segmented Sieve (Learn from Codeverse Channel)
7. Finding prime factorisation of a number in $\log n$ per query (Learn from Codeverse Channel)
8. Euler Totient Function (CodeNCode)
9. Fermat Little Theorem (CodeNCode)

Tougher version of number theory (All from CodeNCode)

1. Finding x^n in $\log(n)$
2. Modular Arithmetic
3. Modular Inverse of a number
3. Modular Exponentiation
4. Chinese Remainder Theorem
5. Factorial Modulo Mod
6. Finding nCr and nPr for queries (constant time)
7. Inclusion Exclusion Principle (combinatorics problems)

1. learn about basic sorting algorithms (bubble, selection, insertion)
2. do problems which are constructive and have a lot of swapping terms in it.

- 3.solve problems related to two pointer approach. (TakeUforward playlist)
- 4.Bit manipulation(left shift, right shift, xor, or, and,set bit,MSB, LSB etc..) (Codeverse)
- 5.Power set of a given array or string using BIT (TakeUforward has a lecture)
- 6.Number of subarrays with XOR as zero(not an algorithm but a must do problem) (TakeUforward)
7. Problems related to greedy algorithm (TakeUforward Playlist)
- 8.Kadane's algorithm (TakeUforward) and problems related to them

Time to learn recursion (Codeverse 3 masterclasses by Striver and TakeUforward Playlist)

After Recursion:

1. Meet in the middle algorithm and problems related to it. (Follow CSES for problems)
- 2.Divide and conquer problems{highly recommended to use codeforces only for this}
3. Next greater element and next smaller element using stack (TakeUforward)
4. problems related to parenthesis.
- 5.largest rectangular area in histogram. (concept is used in a lot of problems) (TakeUforward)
- 6.Problems related to Heap(Priority Queue) {although this gets under the greedy category but by priority queue will help you learn an inbuilt stl}

String algorithms: (Since these are advanced topics, you can read them from where you feel comfortable)

1. Rolling Hash Function on strings {cpalgorithms has a wonderful article written on it} {Spoj or codeforces}
- 2.Rabin Karp Algorithm (cpalgorithms has a wonderful blog on it)
- 3.Prefix Function (cpalgorithms)
- 4.KMP Algorithm
- 5.Z-function
- 6.Manachers' Algorithm (once you have wrapped up the above algorithms, solve a bunch of problems(25-30) on them from different platforms.)

Tree Algorithm :

- 1.Tree/Graph representation
- 2.DFS/BFS Traversal in Graph/Tree
- 3.Basic stuffs(diameter of tree, height of tree, level of tree)
- 4.Euler Tour of Tree(Learn and solve problems)
- 5.Finding LCA using Euler Tour{efficient solution uses segment trees}
- 6.Finding LCA using Binary Lifting. 7.Distance between two nodes.
- 8.Subtree Problems. (SPOJ is highly recommended for trees and codeforces D and E problems also)

Graphs: (TakeUforward Graph series)

- 1.Connected Components.
- 2.Topological Sort.
- 3.Cycle Detection in Graph
- 4.Bipartite Check in graph
- 5.SCC using Kosaraju's algorithm
- 6.Dijkstra's Algorithm

7. Bellman Ford Algorithm
8. Bridges In graphs
9. Articulation Point in a graph
10. Minimum Spanning Tree using Kruskal's Algo
11. Prim's Algorithm
12. 0/1 BFS(a big saviour)
13. Learn Finding Bridges Online(cpalgorithms)

Dynamic Programming:

0. Solve all Standard problems on DP from GFG.. (Refer Aditya Verma youtube channel for learning standard DP)
1. Solve the AtCoder Educational Contests on Dynamic Programming.(all 26)
2. Solve problems from SPOJ(highly recommended,since it doesn't involve any other algorithms)
3. Google dynamic programming practice problem codeforces,u'll get a wonderful blog with a lot of problems on it.
4. Understand how we write recurrence for Digit DP(codeforces blog)(digit dynamic prog)
5. read about DP with Bitmasks and solve problems(hackerearth blog)
6. DP on trees(gfg articles,rachit jains video)
7. SOS DP(cpalgorithm blog)

1. Disjoint Set(cpalgorithms)
2. Offline Queries using Disjoint Set
3. Kruskal's Algorithm using disjoint set

Range Query Algos:

1. Sparse Table(not that imp)
2. Fenwick Tree and Binary Lifting on Fenwick Tree(read about range update trick also)
3. problems on fenwick tree
4. Matrix Exponentiation(problems)
5. Sqrt Decomposition Technique
6. Update and Query Operations
7. Mo's algorithm
8. Mo's algorithm on trees
9. Segment Tree(a must)(Range Queries and point Updates)
10. Lazy propagation on segment trees

Some optional and rare ones:

1. Sprague-Grundy Theorem
2. Flows and Related Problems
3. Heavy Light Decomposition
4. Convex Hull Algorithm
5. FFT/NTT